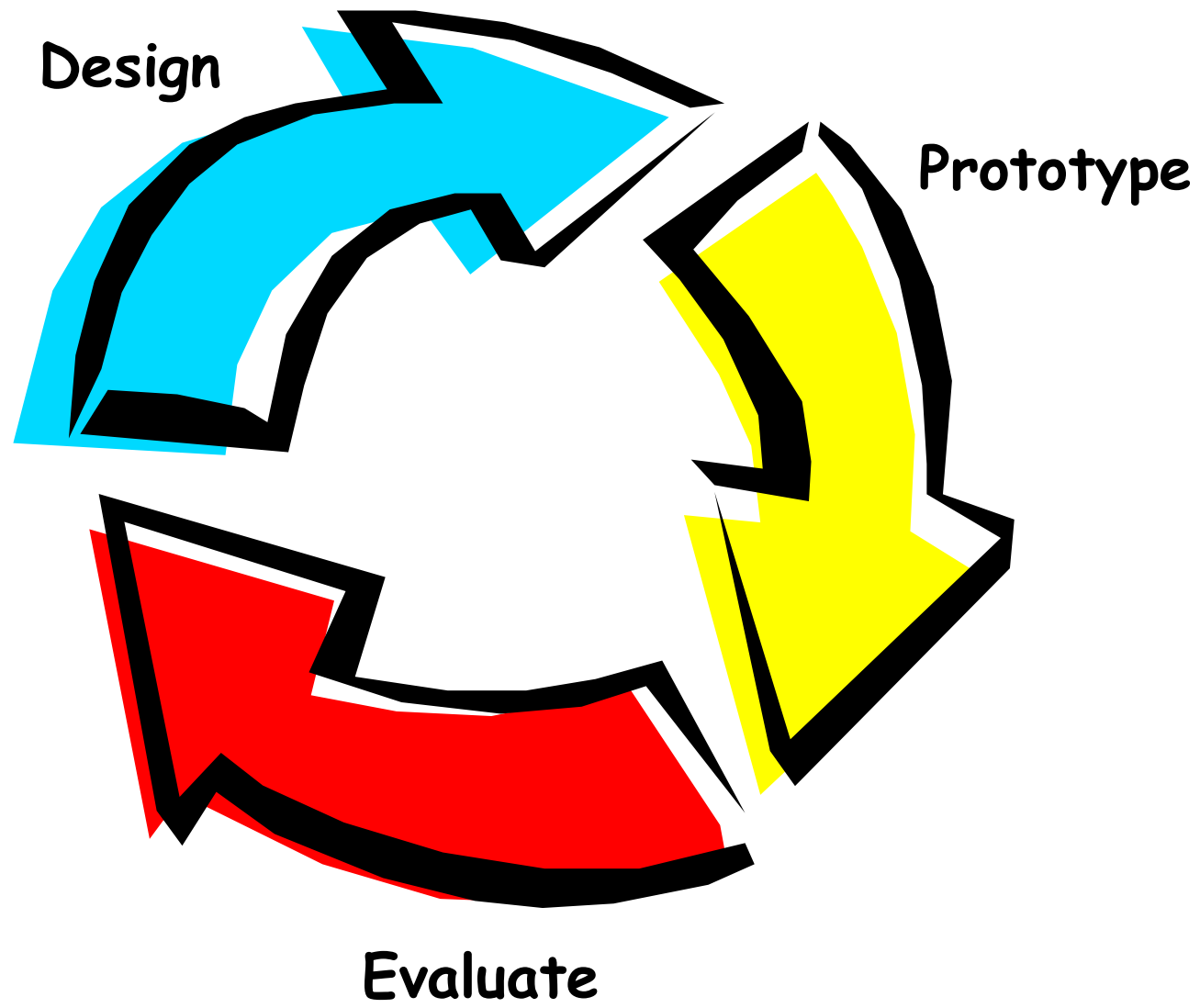


Lecture 9: Qualitative Evaluation

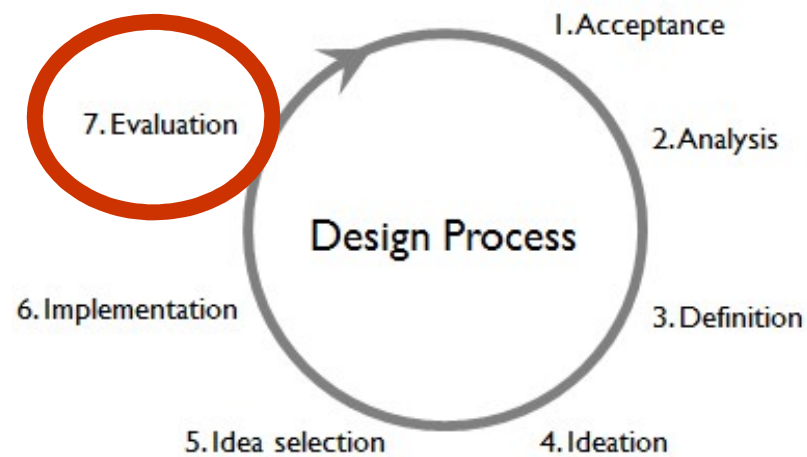
Shengdong Zhao

Acknowledgement:

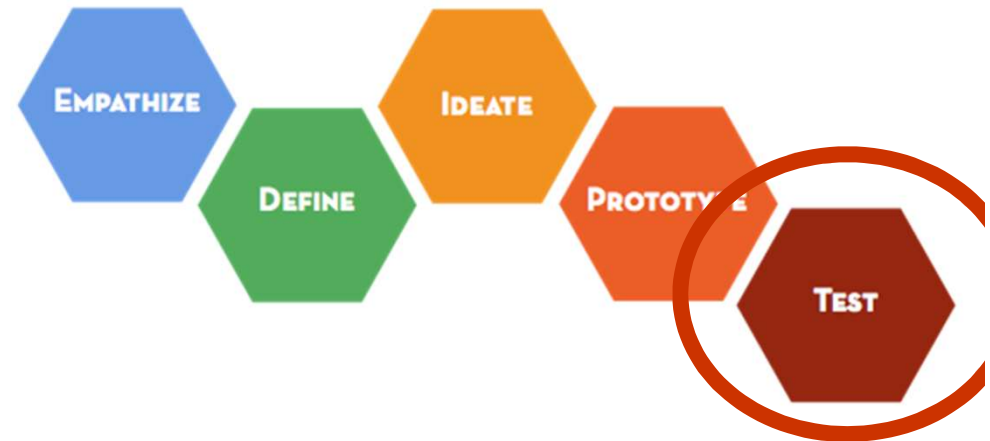
Some material in this lecture is from Saul Greenberg, Maneesh Agrawal, Scott Klemmer, Richard Davis, etc.
Used with permission.



The Design Process



[Koberg & Bagnall]



Design Thinking Workshop

Naturalistic approach

- Observation occurs in realistic setting
 - real life
- Problems
 - hard to arrange and do
 - time consuming
 - may not generalize



Usability engineering approach

Is the test result relevant to the usability of real products in real use outside of lab?

Problems

- non-typical users
- non-typical tasks
- different physical environment
- different social context
 - experimenter vs. boss

•Partial Solution

- use real users
- task-centered system design tasks
- environment similar to real situation



Discount usability evaluation

- Low cost methods to gather usability problems
 - approximate: capture most large and many minor problems

- How?

- Qualitative:
 - observe interactions
 - gather explanations
 - produces description
 - anecdotes, transcripts, problem areas, critical incidents...
- Quantitative*
 - count, log, measure user actions
 - speed, error rate, counts of activities



Qualitative vs. Quantitative



This week

Words



Next week

Numbers

Discount usability evaluation

- Methods

- Inspection
- extracting the conceptual model
- direct observation
 - think-aloud
 - constructive interaction
 - Retrospective Think Aloud
- query techniques (interviews and questionnaires)
- continuous evaluation (user feedback and field studies)



Inspection

- Designer tries the system (or prototype)
 - does the system “feel right”?
 - benefits
 - catch major problems early
 - problems
 - not reliable
 - not valid
 - intuitions can be wrong
- Inspection methods
 - task centered walkthroughs
 - heuristic evaluation



Heuristic Evaluation

Usability Heuristics

“Rules of thumb” describing features of usable systems

- Can be used as design principles
- Can be used to evaluate a design

Example: *Minimize users’ memory load*

Pros and cons

- Easy and inexpensive
 - No need users
 - Catch many design flaws
- More difficult than it seems
 - Not a simple checklist
 - Cannot assess how well the interface will address user goals

Heuristic Evaluation

Developed by Jakob Nielsen (1994)



Original Heuristics

- HI-1:** Simple and natural dialog
- HI-2:** Speak the users' language
- HI-3:** Minimize users' memory load
- HI-4:** Consistency
- HI-5:** Feedback
- HI-6:** Clearly marked exits
- HI-7:** Shortcuts
- HI-8:** Precise & constructive error messages
- HI-9:** Prevent errors
- HI-10:** Help and documentation

Revised Heuristics

Also developed by Nielsen.

- Based on factor analysis of 249 usability problems
- A prioritized, independent set of heuristics

Revised Heuristics

H2-1: Visibility of system status

H2-2: Match system and real world

H2-3: User control and freedom

H2-4: Consistency and standards

H2-5: Error prevention

H2-6: Recognition rather than recall

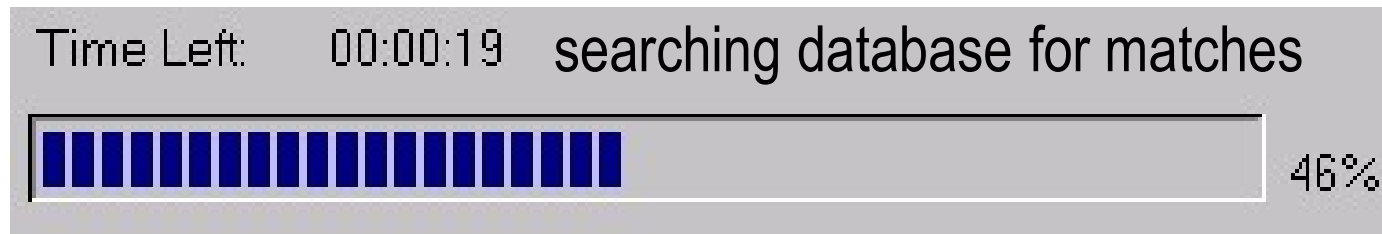
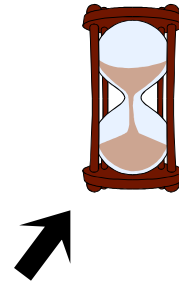
H2-7: Flexibility and efficiency of use

H2-8: Aesthetic and minimalist design

H2-9: Help users recognize, diagnose and recover from errors

H2-10: Help and documentation

Heuristic: Visibility (Feedback)



H2-1: Visibility of system status

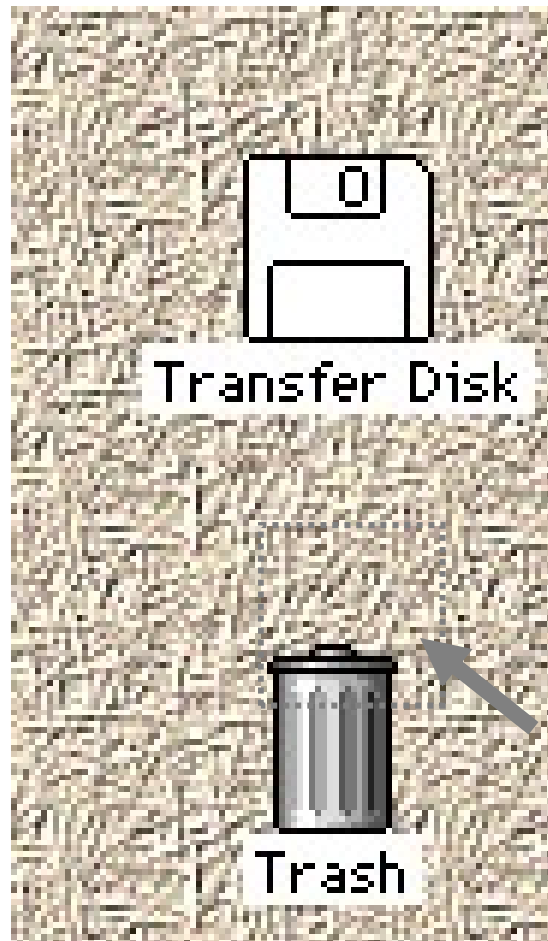
Heuristic: Visibility (Feedback)

Users should always be aware!



Feedback: Toolbar, cursor, ink

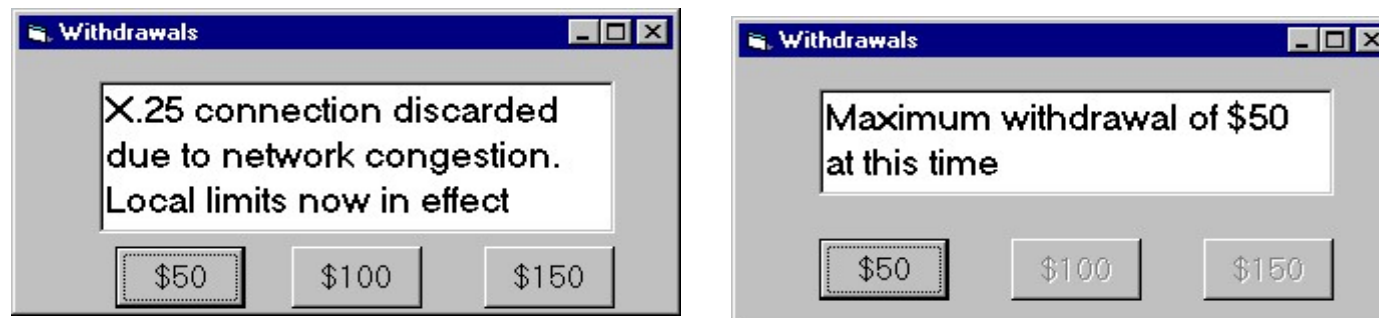
Heuristics (H2-2): Match System & World



Heuristics: Match System & World

Speak users' language

- Withdrawing money at ATM



- Use meaningful mnemonics, icons and abbreviations



Heuristics (2-3) : Control & Freedom



“Exits” for mistaken choices, undo, redo
Don’t force down fixed paths ...

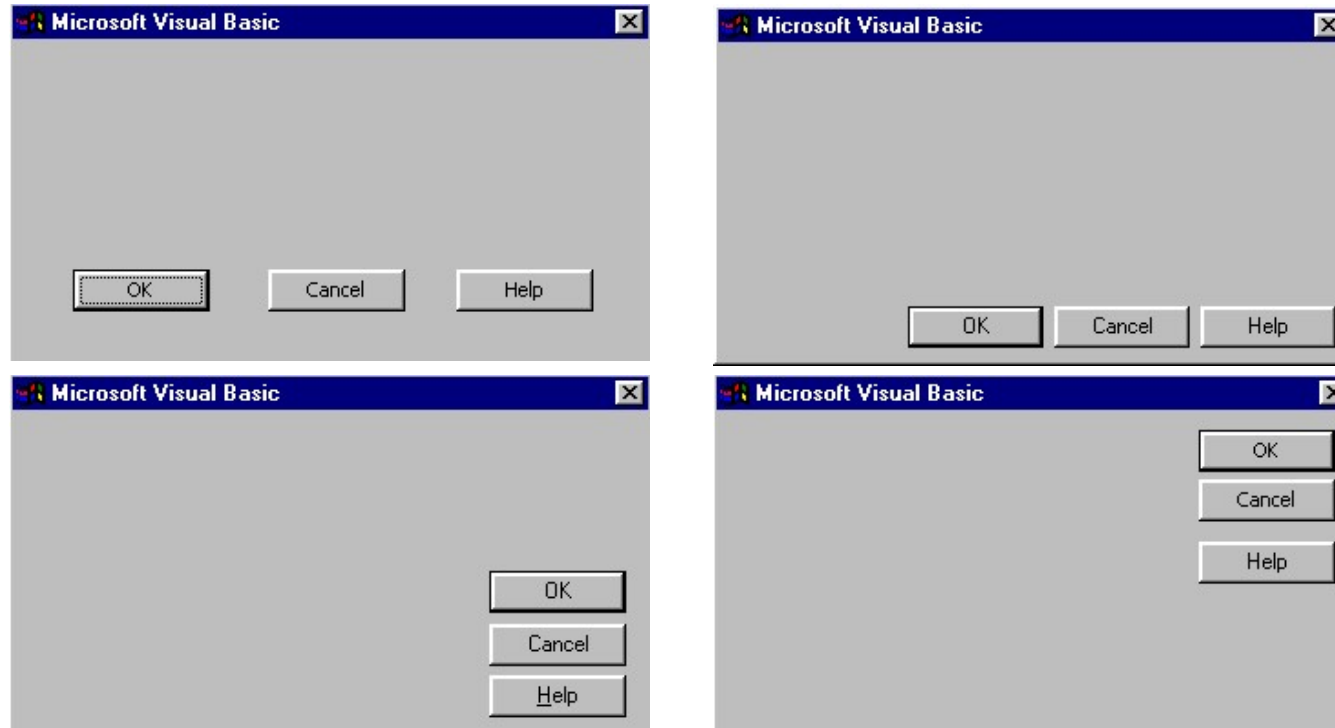
Heuristics: Control & Freedom

- Mark exits: Users don't like to be trapped!



- Strategies
 - Cancel button (or Esc key) for dialog
 - Make the cancel button responsive!
 - Universal undo

Heuristics: Consistency



H2-4: Consistency and standards

Heuristics: Errors and Memory



H2-5: Error prevention

H2-6: Recognition rather than recall

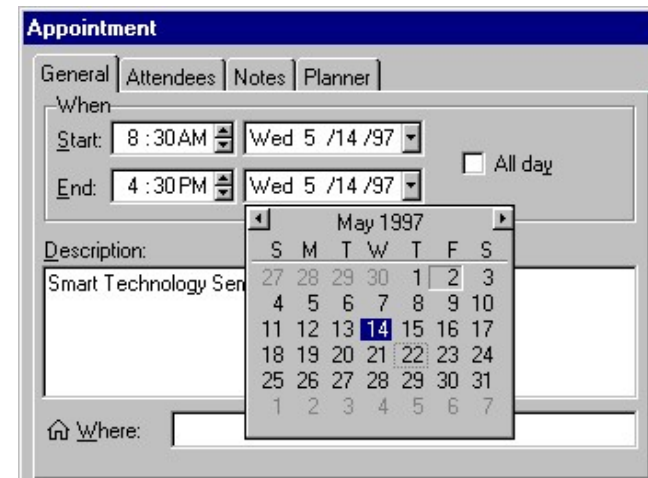
- Make objects, actions, options, & directions visible or easily retrievable

Heuristic: Errors and Memory

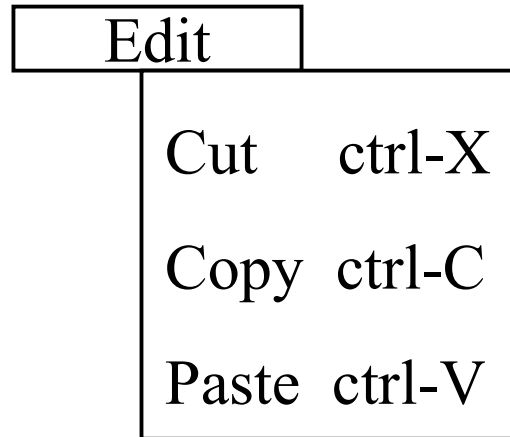
- Promote recognition over recall
 - Recognition is easier than recall



- Describe expected input clearly
 - Don't allow for incorrect input



Heuristics: Flexibility



H2-7: Flexibility and efficiency of use

- Accelerators for experts (e.g., gestures, shortcuts)
- Allow users to tailor frequent actions (e.g., macros)

Heuristics: Aesthetics

Form Title -- (appears above URL in most browsers and is used by w/w/w search)		Background Color:
Q&D Software Development Order Desk		FFFBF0
Form Heading -- (appears at top of Web page in bold type)		Text Color:
Q&D Software Development Order Desk ☒ Center		000080
E-Mail responses to (will not appear on)	Alternate (for mailto forms only)	Background Graphic
dversch@q-d.com		
Text to appear in Submit button	Text to appear in Reset button	☐ Mailto
Send Order	Clear Form	☒ CGI
Scrolling Status Bar Message (max length = 200 characters)		
****WebMania 1.5b with Image Map Wizard is here!!****		
<< Prev Tab		Next Tab >>

H2-8: Aesthetic and minimalist design

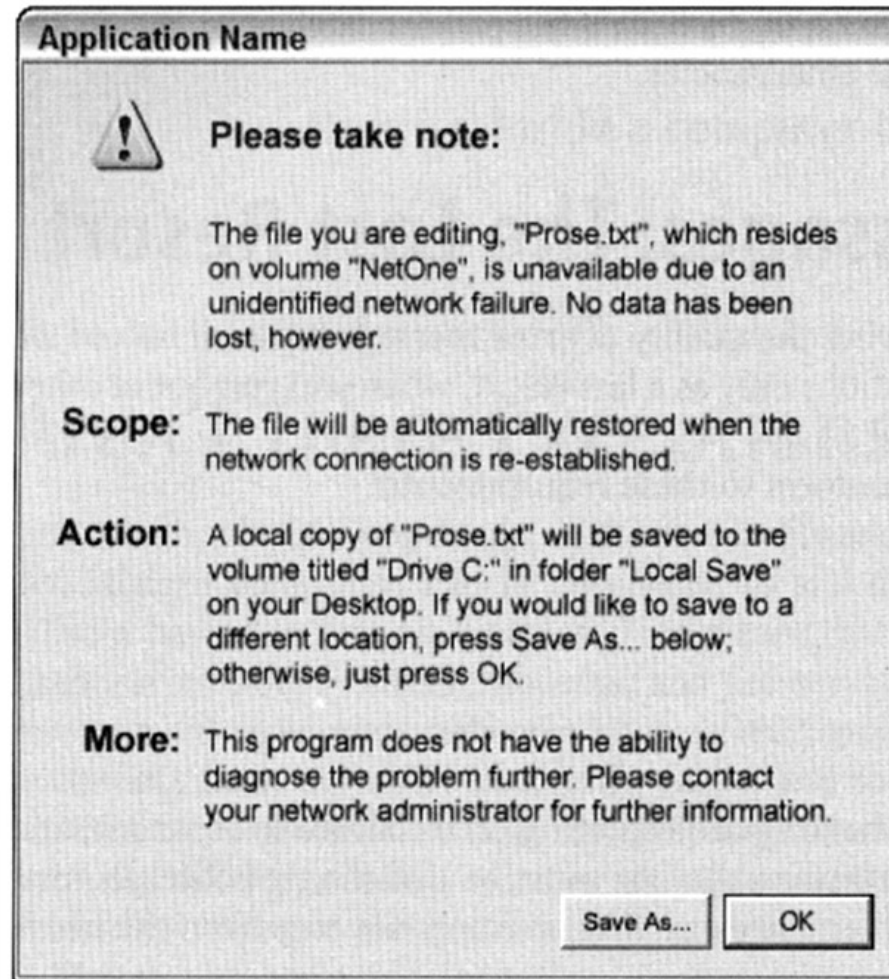
- No irrelevant information in dialogues

Heuristics: Help Users



H2-9: Help users recognize, diagnose, and recover from errors

Good Error Messages

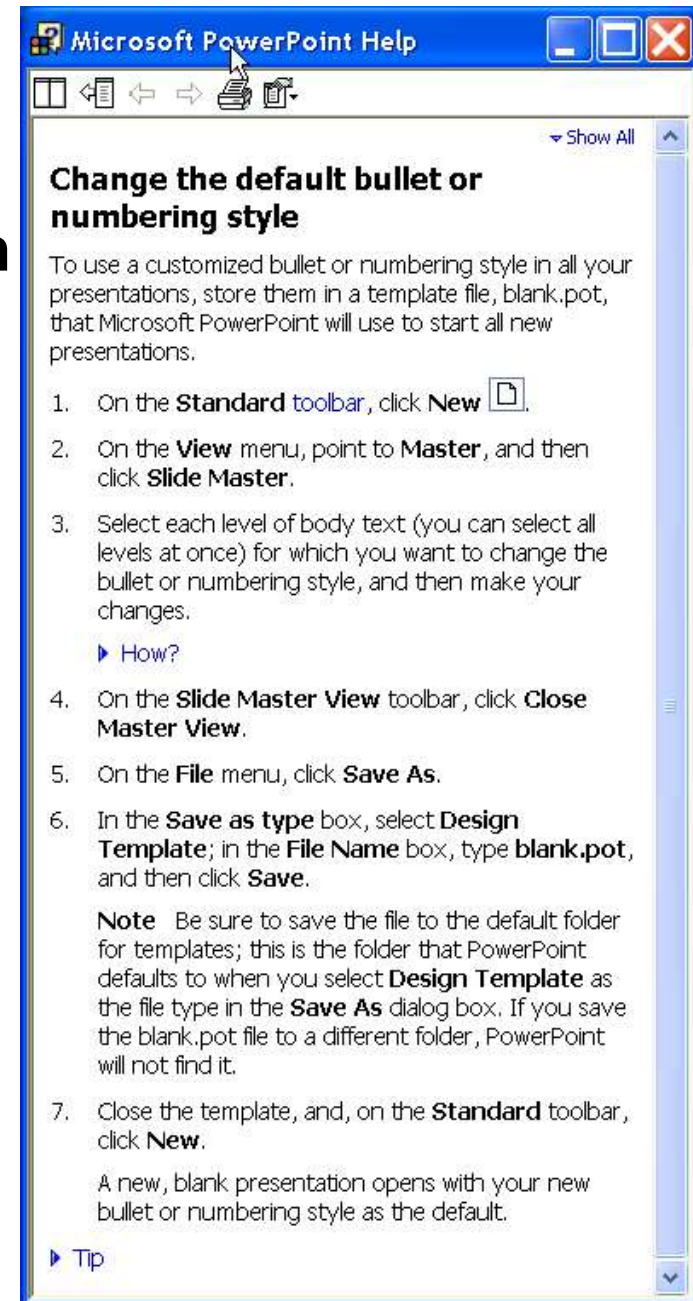


From Cooper's "About Face 2.0"

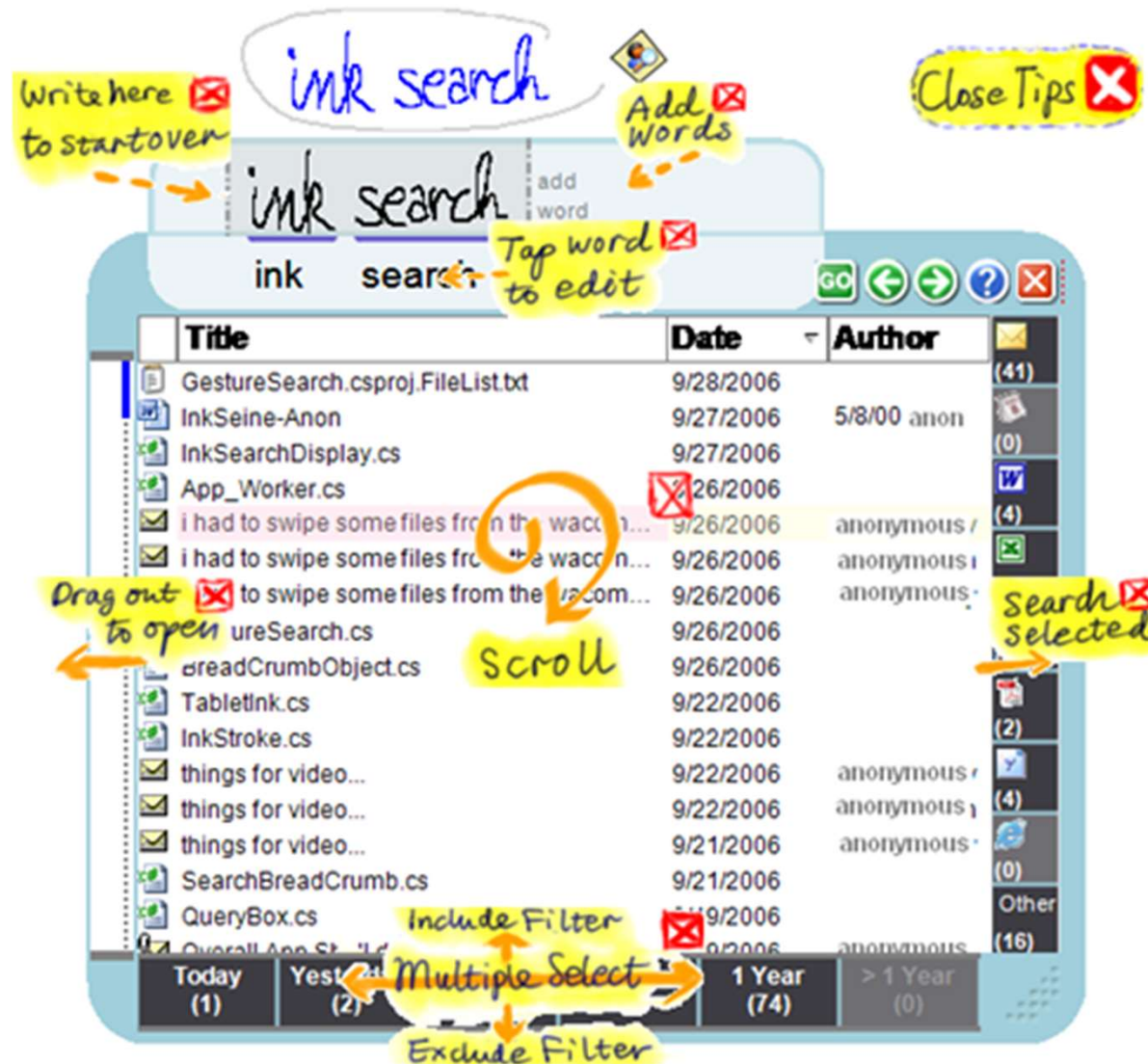
Heuristics: Docs

H2-10: Help and documentation

- Easy to search
- Focused on the user's task
- List concrete steps to carry out
- Not too long



Heuristics: Docs



The Process of Heuristic Evaluation

Phases of Heuristic Eval. (1-2)

1) Pre-evaluation training

2) Evaluation

- Individuals evaluate interface then aggregate results
- Work in 2 passes
 - Overview -> Details
- Each evaluator produces list of problems

Phases of Heuristic Eval. (3-4)

3) Severity rating

- Cosmetic << minor << major << catastrophic

4) Debriefing

- Discuss outcome
- Suggest solutions
- Assess difficulty to fix

Examples

Can't copy info from one window to another

- Violates “User control and freedom” (H2-3)
- Violates “Recognition rather than recall” (H2-7)
- Violates “Flexibility and efficiency of use” (H2-8)
- Fix: allow copying

Typography uses mix of upper/lower case formats and fonts

- Violates “Consistency and standards” (H2-4)
- Slows users down
- Fix: pick a single format for entire interface
- Probably wouldn't be found by user testing

Severity Rating

Used to allocate resources to fix problems

Estimates of need for more usability efforts

Combination of

- Frequency
- Impact
- Persistence (one time or repeating)

Should be calculated after all evaluations are in

Should be done independently by all judges

Levels of Severity

0 - don't agree that this is a usability problem

1 - cosmetic problem

2 - minor usability problem

3 - major usability problem; important to fix

4 - usability catastrophe; imperative to fix

Severity Ratings Example

1. [H2-4 Consistency] [Severity 3][Fix 0]

The interface used the string "Save" on the first screen for saving the user's file, but used the string "Write file" on the second screen. Users may be confused by this different terminology for the same function.

Debriefing

- Conduct with evaluators, observers, and development team members
- Discuss general characteristics of UI
- Suggest improvements to address major usability problems
- Development team rates how hard things are to fix
- Make it a brainstorming session
 - Little criticism until end of session

Number of Evaluators

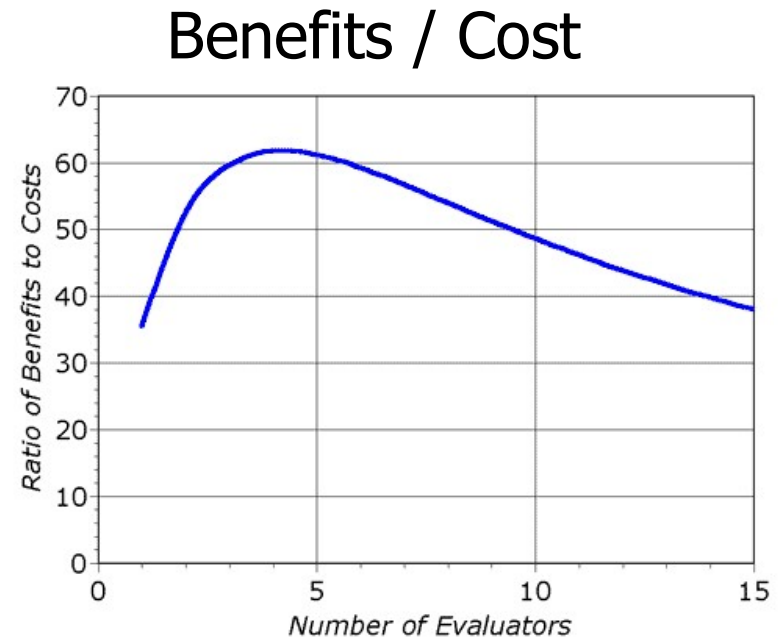
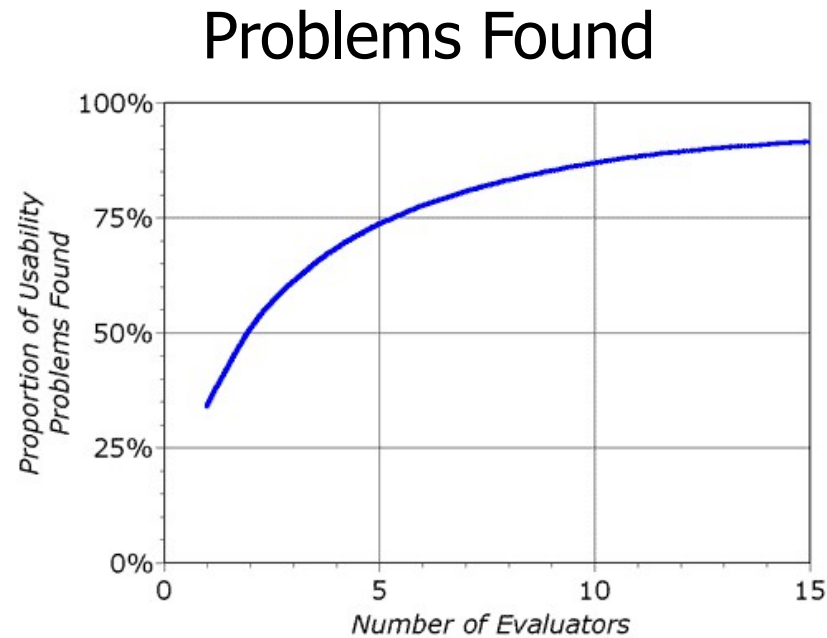
Single evaluator achieves poor results

- Only finds 35% of usability problems
- 5 evaluators find ~ 75% of usability problems
- Why not more evaluators???? 10? 20?
 - Adding evaluators costs more
 - Many evaluators won't find many more problems

But always depends on market for product:

- popular products → high support cost for small bugs

Decreasing Returns



Caveat: Graphs are for a specific example

Summary

- Heuristic evaluation is a discount method
- Have evaluators go through the UI twice
- Have evaluators independently rate severity
- Combine the findings from 3 to 5 evaluators
- Discuss problems with design team
- Cheaper alternative to user testing
 - Finds different problems, so good to alternate

In-class Exercise

Discount usability evaluation

- Methods

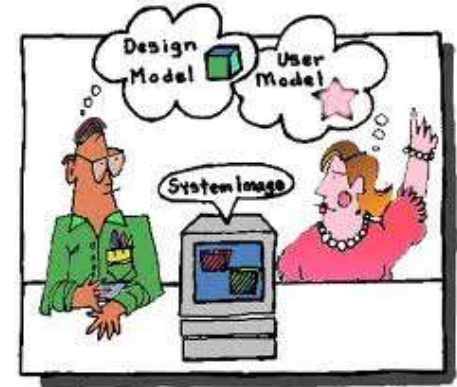
- Inspection
- extracting the conceptual model
- direct observation
 - think-aloud
 - constructive interaction
 - Retrospective Think Aloud
- query techniques (interviews and questionnaires)
- continuous evaluation (user feedback and field studies)



Conceptual model extraction

•How?

- show the user static images of
 - the prototype or screens
- ask the user explain
 - the function of each screen element
 - how they would perform a particular task



•What?

- **Initial conceptual model (first time)**
- **Formative conceptual model (later)**

•Value?

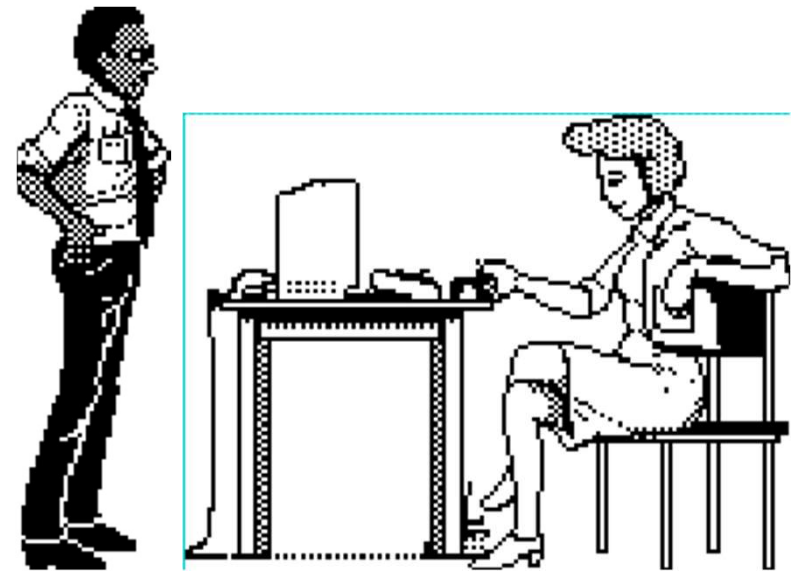
- good for eliciting people's understanding before & after use
- poor for examining system exploration and learning

Direct observations

- Evaluator observes users interacting with system
 - in lab:
 - user asked to complete a set of pre-determined tasks
 - in field:
 - user goes through normal duties
- Value?
 - excellent at identifying gross design/interface problems
 - validity depends on how controlled/contrived the situation is

Simple observation method

- User is given the task
- Evaluator just watches the user
- Problem
 - does not give insight into the user's decision process or attitude



Think aloud method

Users speak their thoughts while doing the task

- gives insight into what the user is thinking
- most widely used evaluation method in industry



Initial Conceptual Model and Think Aloud Exercise

Techniques used:

- Conceptual Model
 - Think Aloud



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DO NOT PULL RECEIVED DOCUMENT FROM FEED
NE PAS TIRER L'ORIGINAL DU DOCUMENT
ATTENDEZ LE DOCUMENT ENVOYÉ PAR L'APPAREIL
N'ARRACHEZ PAS LE DOCUMENT ENVOYÉ PAR L'APPAREIL
NO RETIRE EL DOCUMENT DEL APARATO
NUNQUE SA SORTE SINTE ARRANCA DEL APARATO

Canon
FAX-B320
SIMPLE EXTENSIBLE

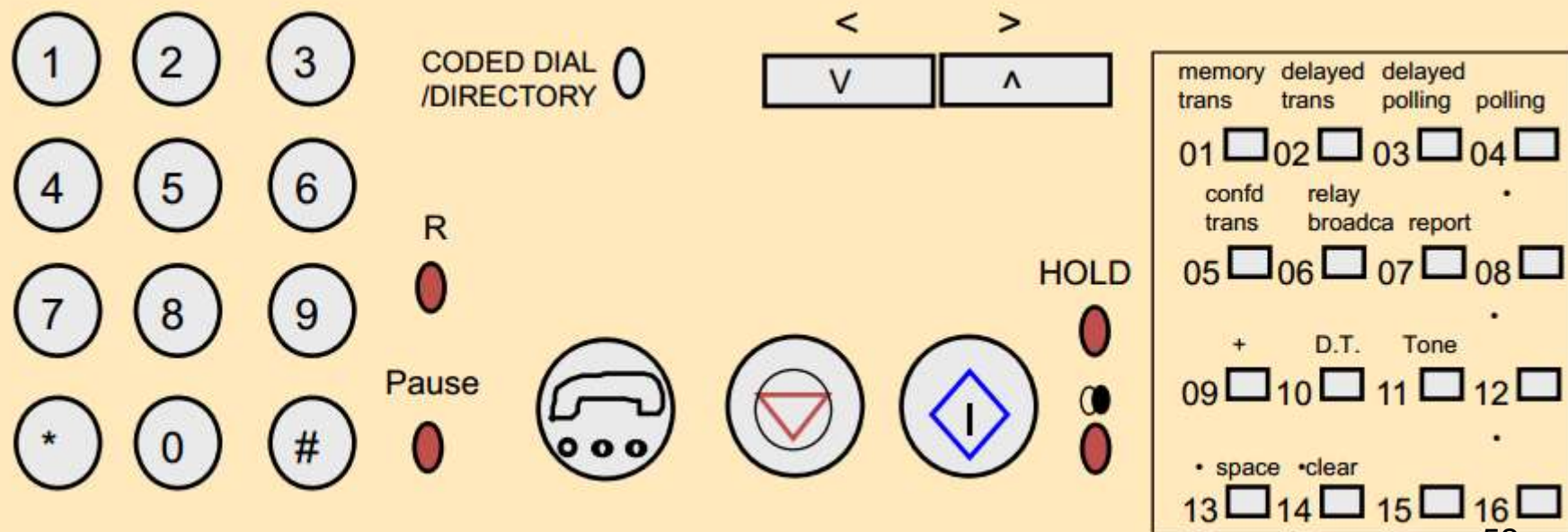
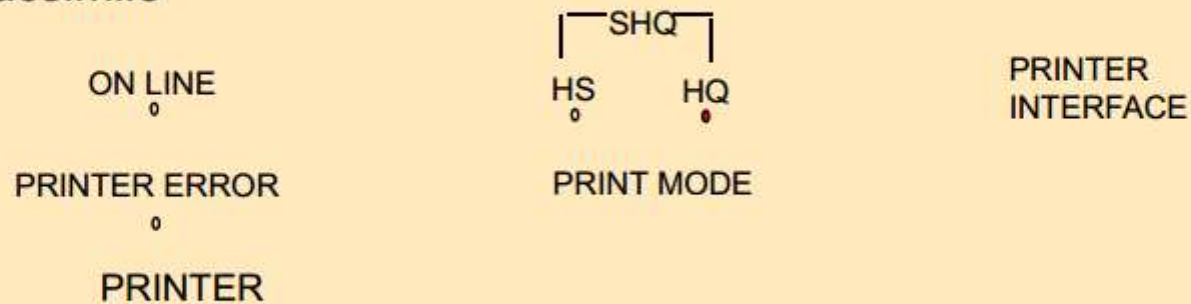


ON LINE
SEND
PRINT
COPY
STOP

PRINTED INTERFACE



Canon Fax-B320 Bubble Jet Facsimile



Think aloud method

Users speak their thoughts while doing the task

- gives insight into what the user is thinking
- most widely used evaluation method in industry

- However:

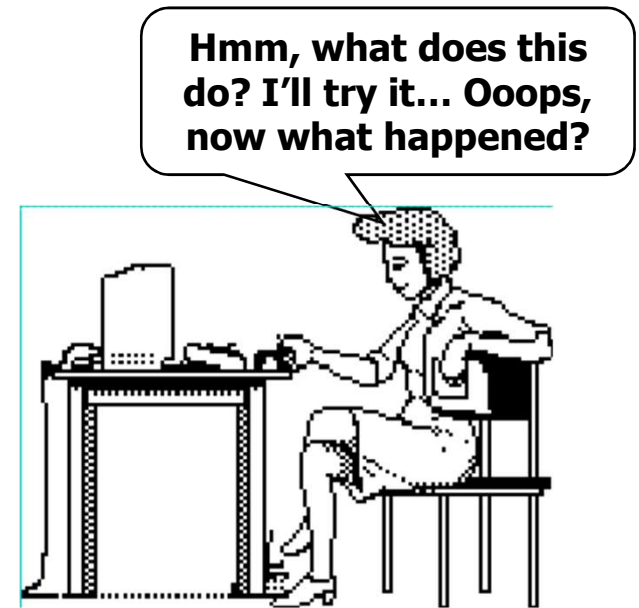
- unnatural (awkward and uncomfortable)
- hard to talk if they are concentrating
- may alter the way users do the task



Problems of Think aloud method

However:

- **unnatural (awkward and uncomfortable)**
- hard to talk if they are concentrating
- may alter the way users do the task



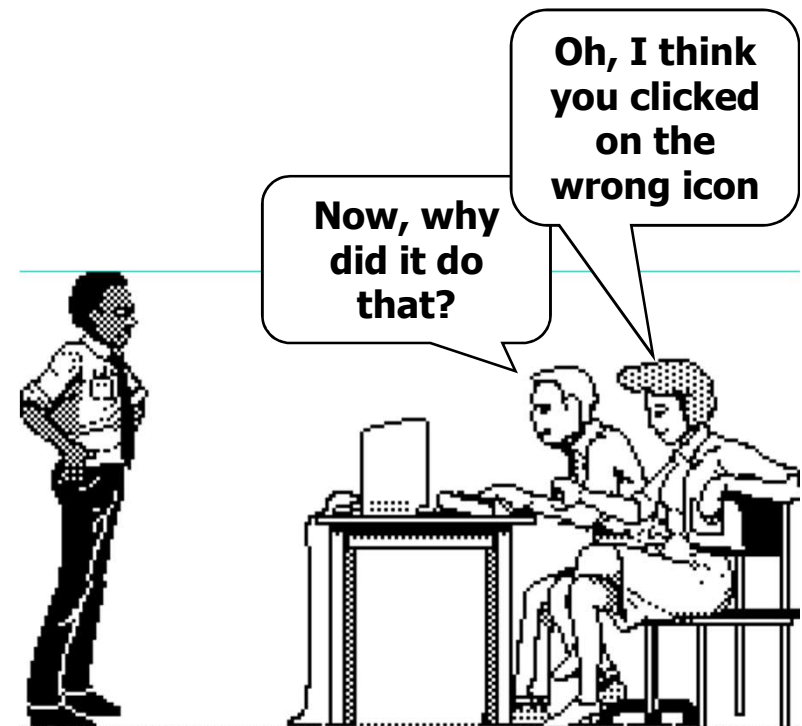
Constructive interaction method

- Two people work together on a task

- monitor their normal conversations

Co-discovery learning

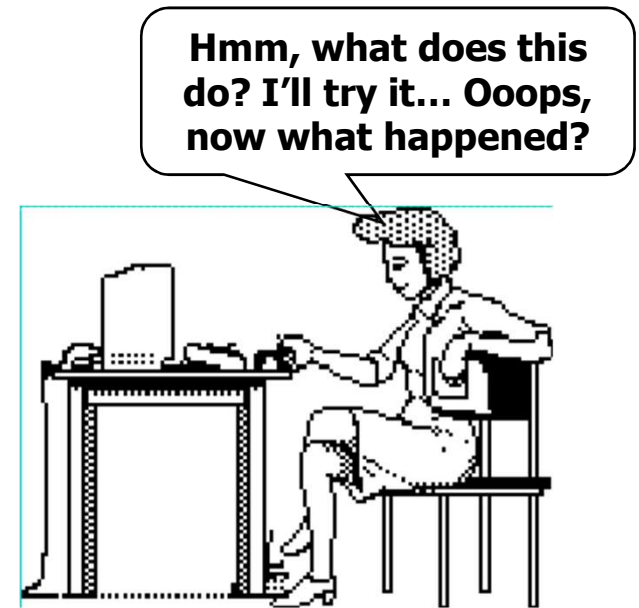
- use semi-knowledgeable “coach” and novice
- only novice uses the interface
 - novice ask questions
 - coach responds
- gives insights into two user groups



Problems of Think aloud method

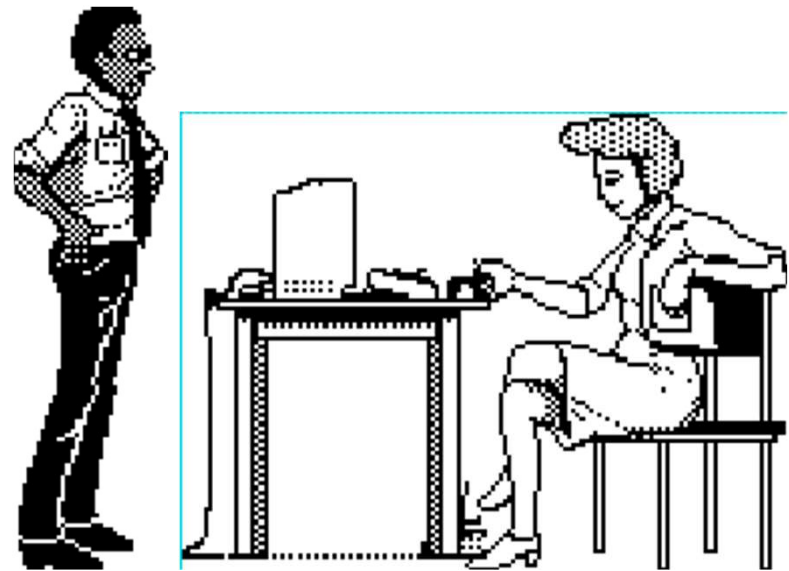
However:

- unnatural (awkward and uncomfortable)
- **hard to talk if they are concentrating**
- **may alter the way users do the task**

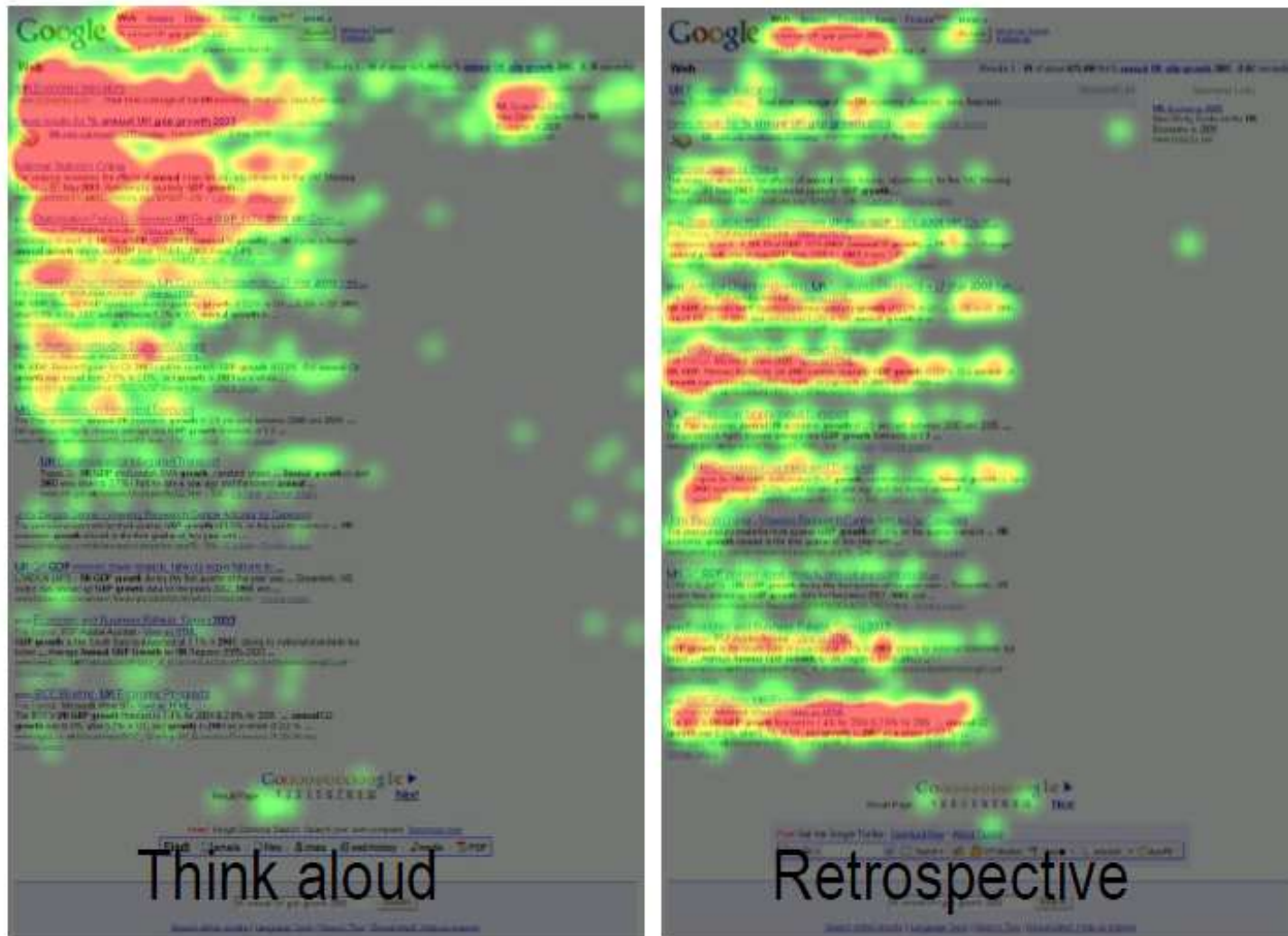


RTA – Retrospective Think Aloud

- Users first complete the task and verbalize after
- Process is observed and recorded with notes
- Benefits
 - Verbalizing on a higher level
 - More relaxed
 - Fabrication not a problem



Comparing Eye-tracking Patterns



Source: Dr L Ball, Lancaster University

What you have learned

- Why do we need evaluation?
- Different stages where usability evaluation applies
- What a usability room look like?
- A number of discounted usability evaluation methods
 - Inspection
 - Initial Conceptual Model
 - Direct observation
 - Think aloud
 - Constructive interaction method
 - RTA (Retrospective Think Aloud)

Steps to Prepare and Conduct a User Study

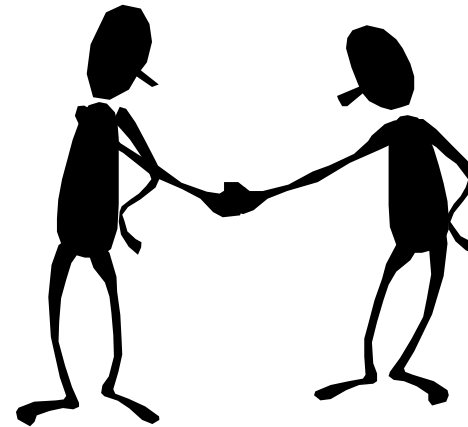
Preparing for a User Test

- Objective: narrow or broad?
- Design the tasks
- Decide on whether to use video/audio
- Choose the setting
- Representative users



User Test

- Roles:
 - Greeter
 - Facilitator: Help users to think aloud...
 - Observers: record “critical incidents”



Critical Incidents

- Critical incidents are unusual or interesting events during the study.
- Most of them are usability problems.
- They may also be moments when the user:
 - got stuck, or
 - suddenly understood something
 - said “that’s cool” etc.

The User Test

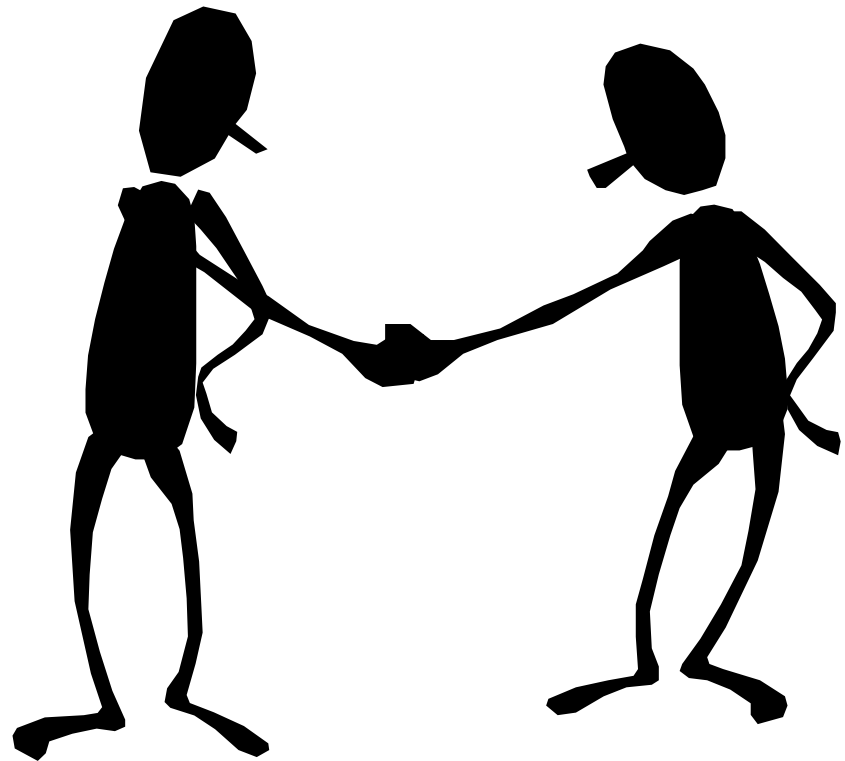
- The actual user test will look something like this:
 - Greet the user
 - Explain the test
 - Collect Demographic Information
 - Get user's signed consent
 - Demo the system
 - Run the test (maybe 1/2 hour)
 - Post-Interview & Questionnaire
 - Debrief

10 Steps to better evaluation

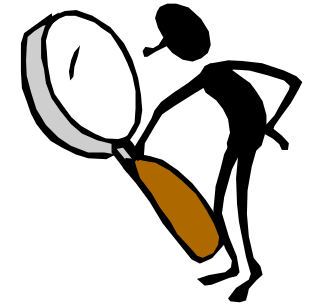
10 steps to better evaluation

1. Introduce yourself

- some background will help relax the subject.



10 steps



2. Describe the purpose of the observation (in general terms), and set the participant at ease
 - You're helping us by trying out this product in its early stages.
 - If you have trouble with some of the tasks, it's the product's fault, not yours. Don't feel bad; that's exactly what we're looking for.

10 steps (contd.)

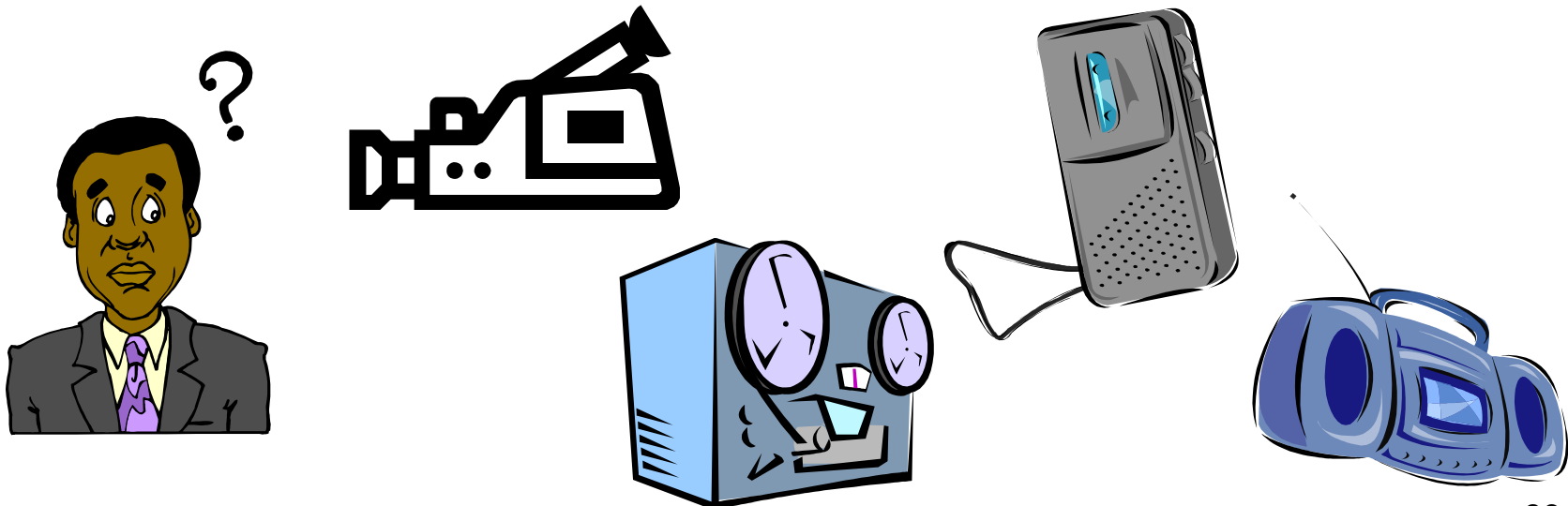
3. Tell the participant that it's okay to quit at any time, e.g.:

- Although I don't know of any reason for this to happen, if you should become uncomfortable or find this test objectionable in any way, you are free to quit at any time.



10 steps (contd.)

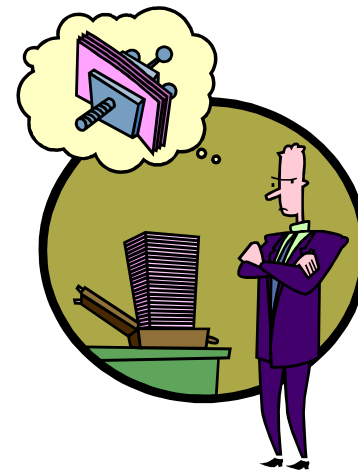
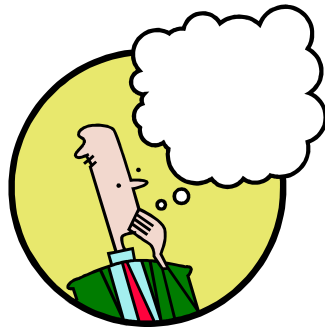
4. Talk about the equipment in the room.
 - Explain the purpose of each piece of equipment (hardware, software, video camera, microphones, etc.) and how it is used in the test.



10 steps (contd.)

5. Explain how to “think aloud.”

- Explain why you want participants to think aloud, and demonstrate how to do it. E.g.:
- We have found that we get a great deal of information from these informal tests if we ask people to think aloud. Would you like me to demonstrate?



10 steps (contd.)

6. Explain that you cannot provide help.



10 steps (contd.)

7. Describe the tasks and introduce the product.

- Explain what the participant should do and in what order. Give the participant written instructions for the tasks.
- However, don't demonstrate what you're trying to test.



10 steps (contd.)

8. Ask if there are any questions before you start; then begin the observation.



10 steps (contd.)

9. Conclude the observation. When the test is over:

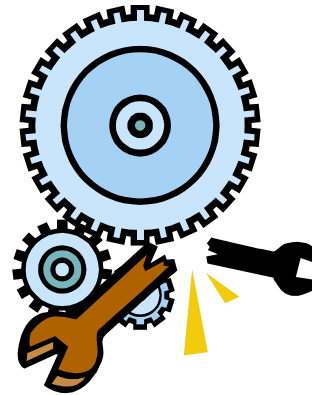
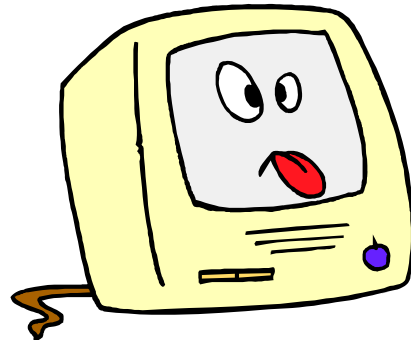
- Explain what you were trying to find.
- Answer any remaining questions.
- Discuss any interesting behaviors you would like the participant to explain.



10 steps (contd.)

10. Use the results.

- When you see participants making mistakes, you should attribute the difficulties to faulty product design, not to the participant.



Using the Results

- Update task analysis and rethink design
 - Rate severity & ease of fixing problems
 - Fix both severe problems & make the easy fixes
- Will thinking aloud give the right answers?
 - Not always
 - If you ask a question, people will always give an answer, even it is has nothing to do with the facts
 - Try to avoid leading questions

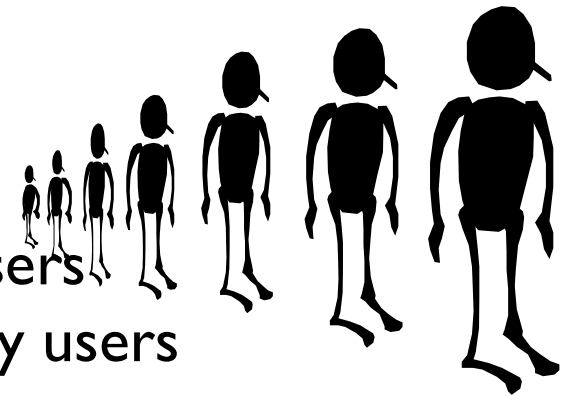
Questions?

High-order summary:

- Follow a loose master-apprentice model
- Observe, but help the user describe what they're doing
- Keep the user at ease

How many users should you observe?

- Problems
 - observing many users is expensive
 - *but* individual differences matter
 - best user 10x faster than slowest
 - best 25% of users ~2x faster than slowest 25%
- Partial solution
 - reasonable number of users with reasonable range
 - big problems usually detected with 3-5 users
 - small problems / fine measures need many users



In-class Exercise

In-class Exercise

- Procedure
 - Greet the user
 - Explain the test
 - (How to be consistent?)
 - Collect demographic information (how?)
 - Get user's signed consent
 - Demo the system
 - (how to be consistent in demoing the system to different users?)
 - Run the test (maybe 1/2 hour)
 - (What test and how to conduct it?)
 - Post-study questionnaire
 - (What type of questions you want to ask?)
 - Debrief
- Work in pairs, and describe briefly what will you do in each step

What we have prepared?

 Practice Session	3/23/2012 11:01 AM	File folder	
 Santa Barbara - USA	3/23/2012 11:01 AM	File folder	
 1-prequestionnaire.docx	4/11/2011 6:37 AM	Microsoft Word D...	108 KB
 2-Practice.docx	4/11/2011 8:34 AM	Microsoft Word D...	105 KB
 3-Manual.docx	4/14/2011 12:34 PM	Microsoft Word D...	87 KB
 4-Sessions.docx	4/14/2011 12:24 PM	Microsoft Word D...	20 KB
 5-Session-additionaltasks.docx	4/14/2011 12:37 PM	Microsoft Word D...	21 KB
 6-Postquestionnaire.docx	4/11/2011 9:17 AM	Microsoft Word D...	130 KB
 Documents to open.docx	4/13/2011 11:57 AM	Microsoft Word D...	13 KB
 Instructor-document.docx	4/13/2011 11:57 AM	Microsoft Word D...	122 KB
 Paris-Itinerary.docx	4/13/2011 11:31 AM	Microsoft Word D...	127 KB

Pre-Study Questionnaire

Experiment·AutoComPaste..... → →Participant number:.....¶

PRE-QUESTIONNAIRE¶

¶
Age:·+_____ → Gender ☐·M·☐·F..... Occupation:_____¶

¶
Are you:··· ☐·Left-handed···· ☐·Right-handed¶

¶
How often do you use a computer a day, overall?¶
☐·less than 1 h··· ☐·1-2 hours···· ☐·3-4 hours··· ☐·5-6 hours··· ☐·more than 6h¶

¶
How familiar are you with text editors?¶
☐·never use them····· ☐·not very familiar····· ☐·familiar····· ☐·expert¶

¶
Which text editors are you the most familiar with (if it applies)?¶

¶
¶

Example Questionnaires

<http://oldwww.acm.org/perlman/question.html>

Acronym	Instrument	Reference	Institution	Example
QUIS	Questionnaire for User Interface Satisfaction	Chin et al, 1988	Maryland	27 questions
PUEU	Perceived Usefulness and Ease of Use	Davis, 1989	IBM	12 questions
NAU	Nielsen's Attributes of Usability	Nielsen, 1993	Bellcore	5 attributes
NHE	Nielsen's Heuristic Evaluation	Nielsen, 1993	Bellcore	10 heuristics
CSUQ	Computer System Usability Questionnaire	Lewis, 1995	IBM	19 questions
ASQ	After Scenario Questionnaire	Lewis, 1995	IBM	3 questions
PHUE	Practical Heuristics for Usability Evaluation	Perlman, 1997	OSU	13 heuristics
PUTQ	Purdue Usability Testing Questionnaire	Lin et al, 1997	Purdue	100 questions
USE	USE Questionnaire	Lund, 2001	Sapient	30 questions

A Simple Questionnaire

			1	2	3	4	5	6	7		NA
1. Learnability		bad	☐	☐	☐	☐	☐	☐	☐	good	☐
2. Efficiency		bad	☐	☐	☐	☐	☐	☐	☐	good	☐
3. Memorability		bad	☐	☐	☐	☐	☐	☐	☐	good	☐
4. Errors (Accuracy)		bad	☐	☐	☐	☐	☐	☐	☐	good	☐
5. Subjective Satisfaction		bad	☐	☐	☐	☐	☐	☐	☐	good	☐
			1	2	3	4	5	6	7		NA

A More Comprehensive One

PERCEIVED USEFULNESS			1	2	3	4	5	6	7		NA
1. Using the system in my job would enable me to accomplish tasks more quickly		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
2. Using the system would improve my job performance		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
3. Using the system in my job would increase my productivity		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
4. Using the system would enhance my effectiveness on the job		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
5. Using the system would make it easier to do my job		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
6. I would find the system useful in my job		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
PERCEIVED EASE OF USE			1	2	3	4	5	6	7		NA
7. Learning to operate the system would be easy for me		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
8. I would find it easy to get the system to do what I want it to do		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
9. My interaction with the system would be clear and understandable		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
10. I would find the system to be flexible to interact with		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
11. It would be easy for me to become skillful at using the system		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
12. I would find the system easy to use		unlikely	☐	☐	☐	☐	☐	☐	☐	likely	☐
			1	2	3	4	5	6	7		NA

Even More Comprehensive

OVERALL REACTION TO THE SOFTWARE			0	1	2	3	4	5	6	7	8	9		NA
1.		terrible	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	wonderful	☐
2.		difficult	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	easy	☐
3.	SCREEN			0	1	2	3	4	5	6	7	8	9	NA
4.	7.	Reading characters on the screen	hard	☐	☐	☐	☐	☐	☐	☐	☐	☐	easy	☐
5.	8.	Highlighting simplifies task	not at all	☐	☐	☐	☐	☐	☐	☐	☐	☐	very much	☐
6.	TERMINOLOGY AND SYSTEM INFORMATION			0	1	2	3	4	5	6	7	8	9	NA
11.	11.	Use of terms throughout system	inconsistent	☐	☐	☐	☐	☐	☐	☐	☐	☐	consistent	☐
	12.	Terminology related to task	never	☐	☐	☐	☐	☐	☐	☐	☐	☐	always	☒
13.	LEARNING			0	1	2	3	4	5	6	7	8	9	NA
14.	17.	Learning to operate the system	difficult	☐	☐	☐	☐	☐	☐	☐	☐	☐	easy	☐
15.	18.	Exploring new features by trial and error	difficult	☐	☐	☐	☐	☐	☐	☐	☐	☐	easy	☐
16.	SYSTEM CAPABILITIES			0	1	2	3	4	5	6	7	8	9	NA
22.	23.	System speed	too slow	☐	☐	☐	☐	☐	☐	☐	☐	☐	fast enough	☐
23.	24.	System reliability	unreliable	☐	☐	☐	☐	☐	☐	☐	☐	☐	reliable	☐
24.	25.	System tends to be	noisy	☐	☐	☐	☐	☐	☐	☐	☐	☐	quiet	☒
	26.	Correcting your mistakes	difficult	☐	☐	☐	☐	☐	☐	☐	☐	☐	easy	☐
	27.	Designed for all levels of users	never	☐	☐	☐	☐	☐	☐	☐	☐	☐	always	☐

Next Time

- Quantitative Evaluation